

COMP90077
Lecture 5
Matchings and Vertex Covers

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Caution

- Slides are designed as presentation aid to go along with lecture, not as study material
- Slides not designed to make sense on their own

Today

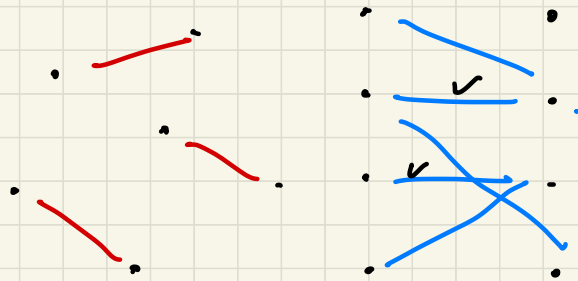
- Graph problems: Matchings and Vertex Covers
 - Greedy-ish (can revoke earlier decisions)
- Warm up to Linear Programming concepts
 - Duality
 - Certifying optimality
- How I approach problems using previous concepts

Matchings

- Does anyone know what a matching is?
- Many variants
- Bipartite graph vs general graph
- Maximize total edge weight
- Minimize total edge weight among perfect matchings

Matchings

Graph $G = (V, E)$

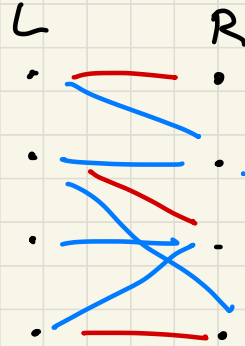


$F \subseteq E$: matching if every vertex incident to at most one edge in F

F is maximal matching if $F \cup \{e\}$ is not a matching for every $e \notin F$

maximum = largest over all matchings.

Bipartite $V = L \cup R$



— Matching
— Not matching.

Matchings: Applications

- Bipartite graphs
 - Job scheduling: Machines – Jobs
 - Uber: Drivers – Riders
 - Kidney exchange: Donor – Recipient
 - Adwords: Advertisers – Impressions
- General graphs
 - Pairing up students for assignments

Keep It Super Simple

- Start with the simplest possible version
 - Steady stream of dopamine/confidence boosts 
 - No external factors to blame: software, hardware
 - Easy to internalize lack of progress
- Bipartite
- Unweighted
 - Maximize number of edges
- Any ideas on how to approach this problem?
 - Not asking for algorithms
 - Asking for thinking process
 - Invitation to think aloud together 

Keep It Super Simple: Pattern Matching

- Does problem fall into one of the previous types?
- What are the previous types?

Matroid Intersection

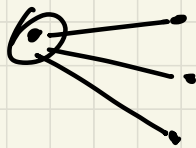
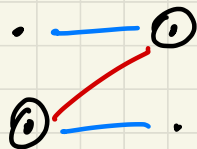
- Ground set = Edges
- Left Matroid: F is independent if at most one edge per left vertex
- Right Matroid: F is independent if at most one edge per right vertex
- F is matching if and only if F is independent for both
- Maximum Matroid Intersection: Find largest common independent set

Keep It Super Simple: What about Greedy?

- What is the Greedy algorithm for this problem?
- Start with empty set
- Add edge if possible
- Important questions about algorithms:
 - Is the algorithm **optimal**? If not, what is the **counterexample**?
 - **Approximation ratio** of the algorithm?
 - Is the **analysis tight**? If yes, what is the **tight example**?
 - Tight example = example on which the ratio matches the analysis

Thus Greedy is $\frac{1}{2}$ -approximation for unweighted

Pf



max matching
 $\cdot \underline{e_1} \cdot \quad v_i \text{ covers } e_i$
 $\cdot \underline{e_2} \cdot \quad v_i \neq v_j$
 $\cdot \underline{e_3} \cdot$

$M = \text{Greedy matching}$

$M^* = \text{Maximum matching}$

Let C be endpoints of M

Every edge in G is incident to some

vertex in C because M is maximal.

C^* is optimal
vertex cover

C is a vertex cover. $|M^*| \leq |C| = 2|M| \leq 2|C^*|$

Lemma $\forall$ matching N , vertex cover D $|N| \leq |D|$

Defn (Vertex Cover) Graph $G = (V, E)$

$D \subseteq V$ is a vertex cover

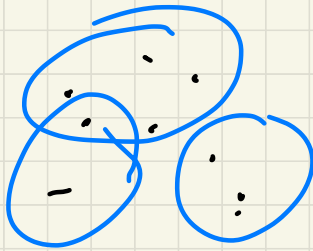
if for every edge $(u, v) \in E$,

either u or v is in D

e.g. V is a vertex cover.

Minimum Vertex Cover: Find smallest D

s.t. D is a vertex cover



Set $S_v = \{(u, v) : u \text{ is a neighbor of } v\}$

Set Cover

Elements

Sets

Vertex Cover

Edges

Vertices

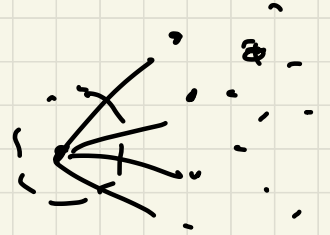
Thm There is a $O(\log n)$ -approximation
for Min Vertex Cover.

Density Greedy for Min VC

$$D = \emptyset$$

while D is not vertex cover:

pick v that is incident to
the most number of uncovered
edges.



Thm There is a 2-approximation for Min VC.

Weak
Duality.

$$\text{Max Matching} \leq \text{Min VC}$$

Max Matching and Min VC are duals.

In bipartite graphs, strong duality.

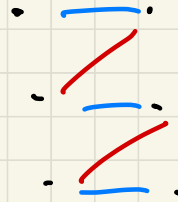
$$\text{Max Matching} = \text{Min VC}$$

Certificate of optimality.

Can check if (M , C)
(matching, vertex cover)

M and C are optimal.
in linear time!

Max Matching in Bipartite Graphs



Matchings M and M'

Symmetric difference

$$M \Delta M'$$

$$\{e: \begin{array}{l} e \in M \text{ and } e \notin M' \\ \text{or} \\ e \in M' \text{ and } e \notin M \end{array}\}$$

$$|M| > |M'|$$